PROBLEM 25

Ijon Tichy's Space Walk

Original statement · Problems.pdf, p. 13

Orbiting Frame + Inertia Forces

A spacecraft S is orbiting a celestial body F. The trajectory of the spacecraft with respect to an inertial reference frame Fxy is given by

$$r = \frac{p}{1 + e\cos\nu},$$

where p and e are known constants and $\nu(t)$ is a known function of time. The space explorer Ijon Tichy needs to perform some extra-vehicular activities during his Seventh Voyage (footnote 1: *The Star Diaries* by Stanislaw Lem). He attaches himself with a massless cord of length L to the spacecraft and goes for a space walk. The cord remains in tension at all times. The astronaut is very light compared to the spacecraft, so his motion does not perturb the motion of the spacecraft. Furthermore, we assume that we can consider him as a point particle of mass m. We define a moving reference frame Sx_1y_1 as depicted in the figure.

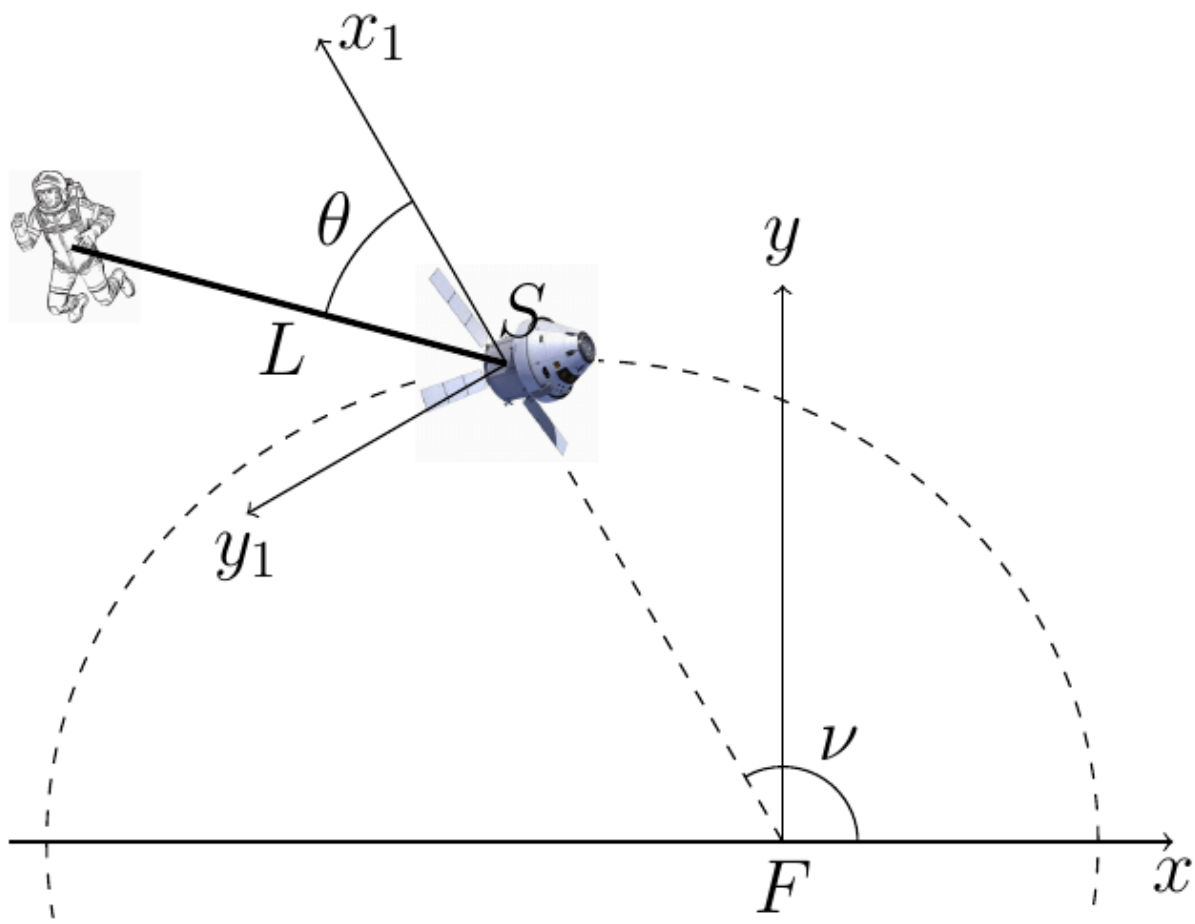


Figure: Spacecraft S on its orbit around F, true anomaly ν, moving axes Sx1 (radial) and Sy1, astronaut on a cord of length L at angle θ from Sx1 (Problems.pdf, p. 13).

(a) Determine the acceleration of the spacecraft with respect to Fxy as a function of time.

(b) Determine the tangent vector to the trajectory of S as a function of time.

(c) Determine the relative velocity and relative acceleration of Ijon Tichy with respect to Sx_1y_1 as a function of θ and its time derivatives.

(d) Determine the fictitious forces acting on Ijon Tichy if we analyze the problem with respect to the frame Sx_1y_1.

The length L is very small compared to the distance between F and S. This allows us to take an approximation: in the following, assume that the gravitational acceleration on the astronaut and on the spacecraft are exactly the same vector (magnitude and direction) as a function of time. (In actual problems the small differences in gravity at the position of the astronaut and the spacecraft can give rise to small but important effects, and this approximation is usually not taken for that reason.)

(e) Write down one ordinary differential equation that once solved (with given initial conditions) would yield the angle θ as a function of time (you do not need to solve it).

(f) Assuming that you solved the previous equation, and therefore knew $\theta(t)$, determine the value of the tension on the cord as a function of time.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a spacecraft S moves on the conic $r = p/(1 + e\cos\nu)$ around the body F, with the true anomaly $\nu(t)$ a known function of time. The astronaut P (mass m) is tied to S by a massless cord of length L that is always taut; θ is the angle from the axis Sx_1 to the cord SP (figure of the statement).
- **Hypotheses:** Fxy is inertial. The motion of S is prescribed (the astronaut does not perturb it). The motion is planar, in the orbital plane Fxy. The cord is massless and taut, so it pulls P towards S with a tension $T > 0$. From part (e) on, the gravitational acceleration at P equals the one at S.
- **Why a moving frame:** the cord fixes the distance $SP = L$, so in a frame attached to S the astronaut moves on a circle of radius L and one angle, θ, describes it. The price is that Sx_1y_1 accelerates and rotates, so inertia forces must be added (Notes Eqs. 4.62–4.63).
- **Fixed (inertial) frame:** $S_0 \equiv Fxy = \{F; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, basis $\mathcal{B}_0$, with $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ normal to the orbital plane.
- **Moving frame:** $S_1 \equiv Sx_1y_1 = \{S; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$. From the figure of the statement, Sx_1 lies on the line FS and points away from F, and Sy_1 is obtained from Sx_1 by a $+90^\circ$ turn about $\mathbf{k}$ (it points in the direction of increasing ν); $\mathbf{k}_1 = \mathbf{k}$. Its origin moves on the

orbit and its axes turn with the radius FS.

- **Basis of the cord:** $\{\mathbf{e}_c, \mathbf{e}_\theta, \mathbf{k}\}$, with $\mathbf{e}_c$ along SP (from S to P) and $\mathbf{e}_\theta$ in the direction of increasing θ; deduced in the next section.
- **Degrees of freedom:** the astronaut moves in the plane (2 coordinates) with one constraint, $|\mathbf{SP}| = L$, so CDOF $= 2 - 1 = 1$, with generalized coordinate θ. The spacecraft adds no unknown, because $\nu(t)$ is given.

Symbol	Meaning	SI unit
p	semi-latus rectum of the orbit of S	m
e	eccentricity of the orbit	dimensionless
$\nu(t)$	true anomaly of S (known)	rad
$r(t)$	distance FS	m
L	cord length	m
m	mass of the astronaut	kg
$\theta(t)$	angle from Sx_1 to the cord	rad
$T(t)$	cord tension	N
μ_F	gravitational parameter GM_F of F (used in the official key only)	m^3/s^2

Change of Basis and Position Vector

The polar basis $\mathcal{B}_1$ in the orbital plane

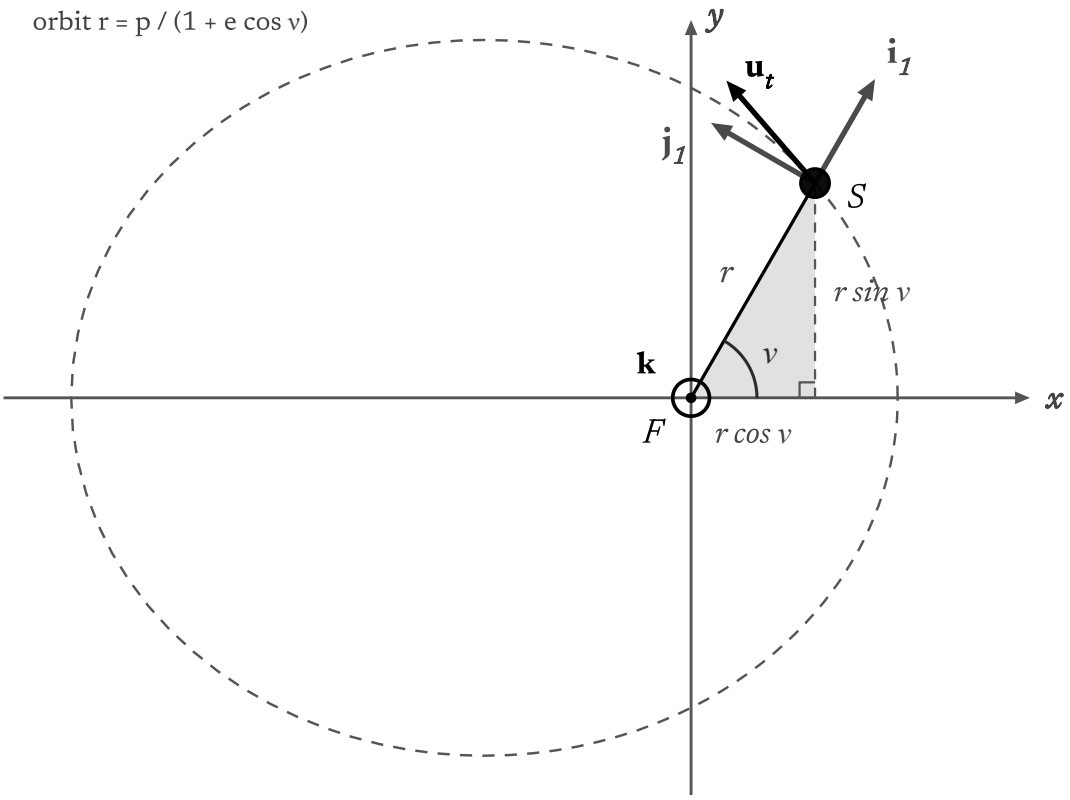


Figure 1. Orbital plane seen from $+\mathbf{k}$ ($\odot$). The shaded right triangle has hypotenuse $|FS| = r$, leg $r\cos\nu$ along $\mathbf{i}$ (adjacent to ν) and leg $r\sin\nu$ along $\mathbf{j}$ (opposite). $\mathbf{i}_1$ is the radial direction and $\mathbf{j}_1$ is $\mathbf{i}_1$ turned $+90^\circ$ (blue); $\mathbf{u}_t$ is the unit tangent of the orbit (orange).

The orbital plane is drawn as seen from $+\mathbf{k}$: $\mathbf{k}$ points out of the paper, towards the reader ($\odot$), $\mathbf{i}$ to the right and $\mathbf{j}$ up (indeed $\mathbf{i}\times\mathbf{j} = \mathbf{k}$: fingers from right to up, thumb towards the reader). The true anomaly ν is measured at F from Fx, counter-clockwise, so it is already the standard polar angle of S and no change of origin of the angle is needed.

Position of the spacecraft. In the shaded right triangle (right angle at the foot of the perpendicular from S to Fx) the hypotenuse is $|FS| = r$, the leg adjacent to ν lies along $\mathbf{i}$ and measures $r\cos\nu$, and the leg opposite to ν lies along $\mathbf{j}$ and measures $r\sin\nu$:

$$\mathbf{r}_0^S = \overrightarrow{FS} = r\cos\nu\,\mathbf{i} + r\sin\nu\,\mathbf{j} = r\,\mathbf{i}_1, \qquad r = \frac{p}{1+e\cos\nu}$$

Radial unit vector. Dividing the same triangle by r gives a triangle with hypotenuse 1, adjacent leg $\cos\nu$ and opposite leg $\sin\nu$: $\mathbf{i}_1 = \cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}$.

Transverse unit vector. $\mathbf{j}_1$ is $\mathbf{i}_1$ turned $+90^\circ$ about $\mathbf{k}$, which in this view (with $\mathbf{k}$ towards the reader) is a counter-clockwise quarter turn: $\mathbf{j}_1 = \mathbf{k} \times \mathbf{i}_1$. With $\mathbf{k} \times \mathbf{i} = \mathbf{j}$ and $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$:

$$\mathbf{j}_1 = \cos\nu\,(\mathbf{k} \times \mathbf{i}) + \sin\nu\,(\mathbf{k} \times \mathbf{j}) = -\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}$$

The minus sign comes from $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$: in the drawing, turning the vertical part $\sin\nu\,\mathbf{j}$ of $\mathbf{i}_1$ a quarter turn counter-clockwise makes it point to the left, along $-\mathbf{i}$. For $0 < \nu < \pi/2$ the tip of $\mathbf{j}_1$ is indeed to the left of the y axis, as drawn. Summary:

$$\mathbf{i}_1 = \cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}, \qquad \mathbf{j}_1 = -\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}, \qquad \mathbf{k}_1 = \mathbf{k}$$

Change-of-basis matrix (columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ written in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\nu & -\sin\nu & 0 \\ \sin\nu & \cos\nu & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1 \cdot (\cos\nu\cos\nu - (-\sin\nu)\sin\nu) = \cos^2\nu + \sin^2\nu = 1$$

The product $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\nu + \sin^2\nu = 1$ (and 1) and off-diagonal entries $\cos\nu\sin\nu - \sin\nu\cos\nu = 0$, so it is the identity.

Angular velocity and acceleration of S_1. S_1 performs a simple rotation of angle $\nu(t)$ about $\mathbf{k}$ (vault note *Topic 5 - Relative Motion*, Sec. 5.5; Slide 16). Check with Poisson's formula: by the chain rule, $\frac{d\mathbf{i}_1}{dt}\big|_0 = \frac{d\cos\nu}{d\nu}\dot\nu\,\mathbf{i} + \frac{d\sin\nu}{d\nu}\dot\nu\,\mathbf{j} = \dot\nu\,(-\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}) = \dot\nu\,\mathbf{j}_1 = \dot\nu\,\mathbf{k} \times \mathbf{i}_1$. Hence

$$\boldsymbol{\omega}_{10} = \dot\nu\,\mathbf{k}, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d(\dot\nu\,\mathbf{k})}{dt}\right|_0 = \ddot\nu\,\mathbf{k}$$

Origin of S_1: $\mathbf{r}_0^S = r\,\mathbf{i}_1$, with $r = p/(1+e\cos\nu)$; $\mathbf{v}_0^S$ and $\mathbf{a}_0^S$ are computed in the kinematics.

The cord in the moving frame

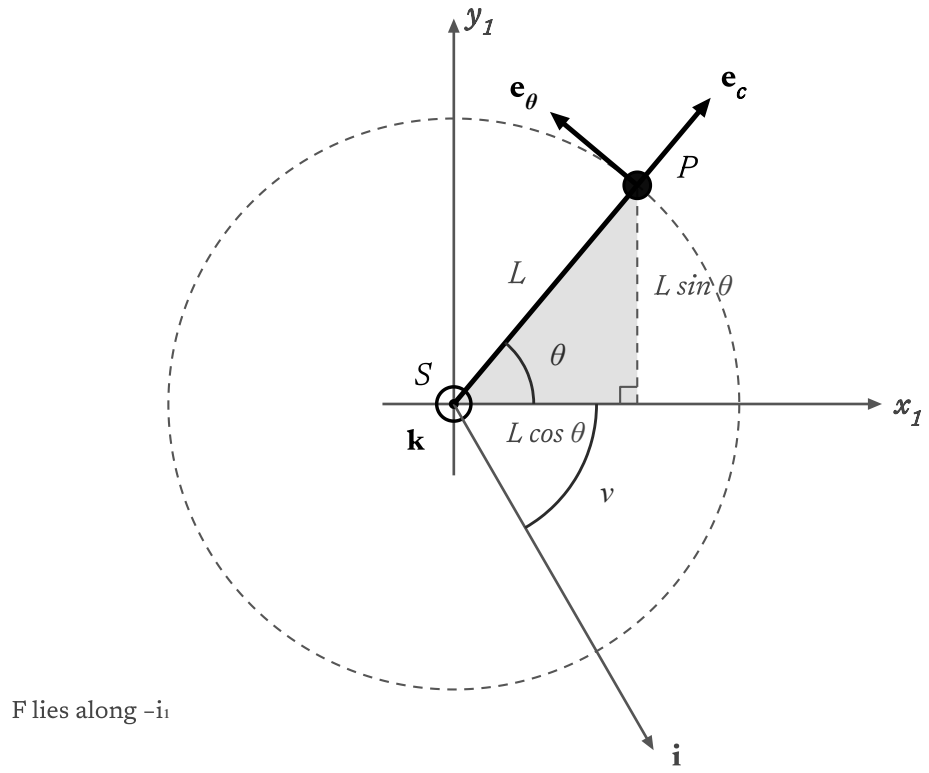


Figure 2. Frame Sx_1y_1 seen from $+\mathbf{k}$ ($\odot$), with $\mathbf{i}_1$ to the right. In the shaded triangle the hypotenuse is the cord $|SP| = L$, the leg adjacent to θ is $L\cos\theta$ (along $\mathbf{i}_1$) and the opposite leg is $L\sin\theta$ (along $\mathbf{j}_1$). $\mathbf{e}_c, \mathbf{e}_\theta$ in orange. The inertial axis $\mathbf{i}$ (grey) is at $-\nu$ from $\mathbf{i}_1$, so the cord makes $\theta + \nu$ with it.

The frame Sx_1y_1 is drawn with $\mathbf{i}_1$ to the right, $\mathbf{j}_1$ up and $\mathbf{k}$ out of the paper ($\odot$); F lies along $-\mathbf{i}_1$, at the distance r. The angle θ is measured from $\mathbf{i}_1$ towards $\mathbf{j}_1$, so it is the standard polar angle of P in $\mathcal{B}_1$. In the shaded right triangle the hypotenuse is the cord, $|SP| = L$; the leg adjacent to θ lies along $\mathbf{i}_1$ and measures $L\cos\theta$; the opposite leg lies along $\mathbf{j}_1$ and measures $L\sin\theta$. Dividing by L:

$$\mathbf{e}_c = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \qquad \mathbf{r}_1^P = \overrightarrow{SP} = L\,\mathbf{e}_c = L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1$$

The transverse vector is the derivative of $\mathbf{e}_c$ with respect to θ ($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$), of length $\sqrt{\sin^2\theta + \cos^2\theta} = 1$; it equals $\mathbf{k} \times \mathbf{e}_c$, the same quarter turn as before:

$$\mathbf{e}_\theta = \frac{d\mathbf{e}_c}{d\theta} = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

$\{\mathbf{e}_c, \mathbf{e}_\theta, \mathbf{k}\}$ is again a rotation of $\mathcal{B}_1$ by θ about $\mathbf{k}$, with determinant $\cos^2\theta + \sin^2\theta = 1$. Products used below: $\mathbf{k}\times\mathbf{e}_c = \mathbf{e}_\theta$ and $\mathbf{k}\times\mathbf{e}_\theta = -\mathbf{e}_c$.

Absolute position of the astronaut. To go from F to P we pass through the spacecraft:

$$\mathbf{r}_0^P = \overrightarrow{FS} + \overrightarrow{SP} = r\,\mathbf{i}_1 + L\,\mathbf{e}_c$$

Angle of the cord in the inertial frame. Substituting $\mathbf{i}_1$ and $\mathbf{j}_1$ in $\mathbf{e}_c$ and using the addition formulas $\cos(\theta+\nu) = \cos\theta\cos\nu - \sin\theta\sin\nu$ and $\sin(\theta+\nu) = \sin\theta\cos\nu + \cos\theta\sin\nu$:

$$\mathbf{e}_c = \cos\theta\,(\cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}) + \sin\theta\,(-\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}) = \cos(\theta+\nu)\,\mathbf{i} + \sin(\theta+\nu)\,\mathbf{j}$$

so the cord makes the angle $\theta+\nu$ with the inertial axis Fx (the grey axis $\mathbf{i}$ of the second figure is at $-\nu$ from $\mathbf{i}_1$). This is used in the checks.

Kinematics

(a) Acceleration of the spacecraft with respect to Fxy

> **Pedagogical Justification:** $\mathbf{r}_0^S = r\,\mathbf{i}_1$ is written in the rotating basis $\mathcal{B}_1$, so its derivative in S_0 follows from the Coriolis theorem (Notes Eq. 3.8, Slide 17): derivative of the components in S_1 plus $\boldsymbol{\omega}_{10}\times(\cdot)$.

Velocity. In S_1 only the component r changes, and $\dot\nu\,\mathbf{k}\times r\,\mathbf{i}_1 = r\dot\nu\,(\mathbf{k}\times\mathbf{i}_1) = r\dot\nu\,\mathbf{j}_1$:

$$\mathbf{v}_0^S = \left.\frac{d(r\,\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times r\,\mathbf{i}_1 = \dot r\,\mathbf{i}_1 + r\dot\nu\,\mathbf{j}_1$$

Acceleration. Apply the theorem again. The derivative of the components in S_1 uses the product rule on $r\dot\nu$, $\frac{d}{dt}(r\dot\nu) = \dot r\dot\nu + r\ddot\nu$:

$$\left.\frac{d\mathbf{v}_0^S}{dt}\right|_1 = \ddot{r}\,\mathbf{i}_1 + (\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

The rotation term, with $\mathbf{k}\times\mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}\times\mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\omega}_{10}\times\mathbf{v}_0^S = \dot{\nu}\,\mathbf{k}\times(\dot{r}\,\mathbf{i}_1 + r\dot{\nu}\,\mathbf{j}_1) = \dot{r}\dot{\nu}\,\mathbf{j}_1 - r\dot{\nu}^2\,\mathbf{i}_1$$

Adding both:

$$\mathbf{a}_0^S = \left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 + (2\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

Derivatives of $r(\nu(t))$. Write $r = p\,D^{-1}$ with $D = 1 + e\cos\nu$. By the chain rule, $\dfrac{dD}{dt} = \dfrac{dD}{d\nu}\dot{\nu} = -e\sin\nu\,\dot{\nu}$, so

$$\dot{r} = \frac{dr}{dD}\frac{dD}{dt} = -p\,D^{-2}\,(-e\sin\nu\,\dot{\nu}) = \frac{p\,e\,\dot{\nu}\sin\nu}{(1+e\cos\nu)^2}$$

Differentiate again with the product rule on $N = pe\,\dot{\nu}\sin\nu$ and D^{-2}: $\dot{N} = pe\left(\ddot{\nu}\sin\nu + \dot{\nu}^2\cos\nu\right)$ (chain rule $\frac{d}{dt}\sin\nu = \cos\nu\,\dot{\nu}$) and $\frac{d}{dt}D^{-2} = -2D^{-3}\dot{D} = 2e\sin\nu\,\dot{\nu}\,D^{-3}$:

$$\ddot{r} = \frac{pe\left(\ddot{\nu}\sin\nu + \dot{\nu}^2\cos\nu\right)}{(1+e\cos\nu)^2} + \frac{2pe^2\dot{\nu}^2\sin^2\nu}{(1+e\cos\nu)^3}$$

Since $\nu(t)$ is known, $r, \dot{r}, \ddot{r}, \dot{\nu}, \ddot{\nu}$ are known functions of time, and so is $\mathbf{a}_0^S$. In $\mathcal{B}_0$ (substituting the columns of $[{}_0R_1]$), $\mathbf{a}_0^S = \left[(\ddot{r} - r\dot{\nu}^2)\cos\nu - (2\dot{r}\dot{\nu} + r\ddot{\nu})\sin\nu\right]\mathbf{i} + \left[(\ddot{r} - r\dot{\nu}^2)\sin\nu + (2\dot{r}\dot{\nu} + r\ddot{\nu})\cos\nu\right]\mathbf{j}$.

(b) Tangent vector to the trajectory of S

> **Pedagogical Justification:** the unit tangent is the velocity divided by the speed, $\mathbf{u}_t = \mathbf{v}_0^S/|\mathbf{v}_0^S|$ (intrinsic frame of Topic 1). Taking $\dot\nu > 0$ (prograde motion, as in the figure) the common factor $r\dot\nu$ can be extracted.

From (a), $\dot r = r\dot\nu\,\dfrac{e\sin\nu}{1+e\cos\nu}$ (because $\dfrac{p}{(1+e\cos\nu)^2} = \dfrac{r}{1+e\cos\nu}$). Then

$$\mathbf{v}_0^S = \frac{r\dot\nu}{1+e\cos\nu}\left[e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1\right]$$

The bracket is the hypotenuse of a right triangle with legs $e\sin\nu$ (along $\mathbf{i}_1$) and $1+e\cos\nu$ (along $\mathbf{j}_1$); its length is $\sqrt{e^2\sin^2\nu + 1 + 2e\cos\nu + e^2\cos^2\nu} = \sqrt{1+2e\cos\nu+e^2}$, hence

$$\mathbf{u}_t(t) = \frac{e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1}{\sqrt{1+2e\cos\nu+e^2}}, \qquad |\mathbf{v}_0^S| = \frac{r\dot\nu\sqrt{1+2e\cos\nu+e^2}}{1+e\cos\nu}$$

with $\nu = \nu(t)$. In $\mathcal{B}_0$, substituting $\mathbf{i}_1$ and $\mathbf{j}_1$, the $\mathbf{i}$ component of the bracket is $e\sin\nu\cos\nu - (1+e\cos\nu)\sin\nu = -\sin\nu$ and the $\mathbf{j}$ component is $e\sin^2\nu + (1+e\cos\nu)\cos\nu = e + \cos\nu$, so $\mathbf{u}_t = \left[-\sin\nu\,\mathbf{i} + (e+\cos\nu)\,\mathbf{j}\right]/\sqrt{1+2e\cos\nu+e^2}$ (the orange vector of the first figure). For a circular orbit ($e=0$) $\mathbf{u}_t = \mathbf{j}_1$, perpendicular to the radius, as expected.

(c) Relative velocity and acceleration of the astronaut in Sx_1y_1

> **Pedagogical Justification:** in S_1 the basis $\mathcal{B}_1$ is constant, so only the components of $\mathbf{r}_1^P = L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1$ are differentiated (Notes Eq. 3.7); L is constant.

Chain rule: $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$, $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -L\dot\theta\sin\theta\,\mathbf{i}_1 + L\dot\theta\cos\theta\,\mathbf{j}_1 = L\dot\theta\,\mathbf{e}_\theta$$

Differentiate again; product rule on $\dot\theta\sin\theta$: $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta^2\cos\theta$, and on $\dot\theta\cos\theta$: $\ddot\theta\cos\theta - \dot\theta^2\sin\theta$:

$$\mathbf{a}_1^P = -L\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\mathbf{i}_1 + L\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\mathbf{j}_1 = L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c$$

This is uniform-radius circular motion: a tangential part $L\ddot{\theta}$ and a centripetal part $L\dot{\theta}^2$ towards S.

Forces and Equations of Motion

(d) Fictitious forces in Sx_1y_1

Pedagogical Justification: in the non-inertial frame S_1, Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), with the four inertia forces of Notes Eq. 4.62 (Slide 28). Each one is computed with the motion of S_1 found in the change-of-basis section.

1. Force due to the acceleration of the origin ($\mathbf{F}_A = -m\mathbf{a}_0^S$), with $\mathbf{a}_0^S$ from (a):

$$\mathbf{F}_A = -m\left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 - m\left(2\dot{r}\dot{\nu} + r\ddot{\nu}\right)\mathbf{j}_1$$

2. Euler force ($\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P$). With $\mathbf{k}\times\mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}\times\mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \ddot{\nu}\,\mathbf{k}\times(L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1) = L\ddot{\nu}\,(\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{i}_1) \;\Rightarrow\; \mathbf{F}_E = mL\ddot{\nu}\,(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1) = -mL\ddot{\nu}\,\mathbf{e}_\theta$$

3. Centrifugal force ($\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)$). First $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = L\dot{\nu}\,\mathbf{e}_\theta$ (same computation as above with $\dot{\nu}$), then $\dot{\nu}\,\mathbf{k}\times L\dot{\nu}\,\mathbf{e}_\theta = L\dot{\nu}^2\,(-\sin\theta\,\mathbf{j}_1 - \cos\theta\,\mathbf{i}_1) = -L\dot{\nu}^2\,\mathbf{e}_c$:

$$\mathbf{F}_C = mL\dot{\nu}^2\,\mathbf{e}_c = mL\dot{\nu}^2\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1)$$

in agreement with $\mathbf{F}_C = m\omega_{10}^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_1^P$ is perpendicular to $\mathbf{k}$.

4. Coriolis force ($\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$). With $\mathbf{k}\times\mathbf{e}_\theta = -\mathbf{e}_c$:

$$\mathbf{F}_{COR} = -2m\,\dot{\nu}\,\mathbf{k} \times L\dot{\theta}\,\mathbf{e}_\theta = 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c = 2mL\dot{\nu}\dot{\theta}\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1)$$

The total inertia force is $\mathbf{F}_i = \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$. Because $\nu(t)$ is in general not uniform ($\ddot{\nu} \neq 0$ on an elliptic orbit), the Euler force is present.

(e) Equation of motion for $\theta(t)$

> **Pedagogical Justification:** the real forces on the astronaut are the cord tension $\mathbf{T} = -T\,\mathbf{e}_c$ (towards S) and gravity $m\mathbf{g}_P$. Projecting Notes Eq. 4.63 on $\mathbf{e}_\theta$ removes the unknown tension, because $\mathbf{e}_c \cdot \mathbf{e}_\theta = 0$.

Gravity hypothesis. S is in free flight around F (its conic orbit is a Keplerian orbit, Topic 4), so its acceleration is the gravitational acceleration at S: $\mathbf{a}_0^S = \mathbf{g}_S$. The statement assumes $\mathbf{g}_P = \mathbf{g}_S$. Hence the weight cancels the force due to the acceleration of the origin exactly:

$$m\mathbf{g}_P + \mathbf{F}_A = m\mathbf{a}_0^S - m\mathbf{a}_0^S = \mathbf{0}$$

Newton's law in S_1:

$$m\left(L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c\right) = -T\,\mathbf{e}_c - mL\ddot{\nu}\,\mathbf{e}_\theta + mL\dot{\nu}^2\,\mathbf{e}_c + 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c$$

Component along $\mathbf{e}_\theta$: $mL\ddot{\theta} = -mL\ddot{\nu}$. (Without the approximation, the same projection gives $L(\ddot{\theta} + \ddot{\nu}) = (\mathbf{g}_P - \mathbf{a}_0^S)\cdot\mathbf{e}_\theta$, the tidal term that the statement neglects.)

Component along $\mathbf{e}_c$:

$$-mL\dot{\theta}^2 = -T + mL\dot{\nu}^2 + 2mL\dot{\nu}\dot{\theta}$$

Answers and Checks

(a) Acceleration of the spacecraft

$\mathbf{a}_0^S = \left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 + \left(2\dot{r}\dot{\nu} + r\ddot{\nu}\right)\mathbf{j}_1$, with $r = p/(1+e\cos\nu)$, $\dot{r} = \dfrac{p\,e\,\dot{\nu}\sin\nu}{(1+e\cos\nu)^2}$ and $\ddot{r}$ as in the kinematics, all known functions of t through $\nu(t)$.

(b) Tangent vector

$$\mathbf{u}_t(t) = \frac{e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1}{\sqrt{1+2e\cos\nu+e^2}} = \frac{-\sin\nu\,\mathbf{i} + (e+\cos\nu)\,\mathbf{j}}{\sqrt{1+2e\cos\nu+e^2}}.$$

(c) Relative velocity and acceleration

$\mathbf{v}_1^P = L\dot{\theta}\,\mathbf{e}_\theta = -L\dot{\theta}\sin\theta\,\mathbf{i}_1 + L\dot{\theta}\cos\theta\,\mathbf{j}_1$ and $\mathbf{a}_1^P = L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c$.

(d) Fictitious forces

$\mathbf{F}_A = -m\mathbf{a}_0^S$, $\mathbf{F}_E = -mL\ddot{\nu}\,\mathbf{e}_\theta$, $\mathbf{F}_C = mL\dot{\nu}^2\,\mathbf{e}_c$, $\mathbf{F}_{COR} = 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c$.

(e) Differential equation for $\theta(t)$

From the $\mathbf{e}_\theta$ component:

$$\ddot{\theta} + \ddot{\nu}(t) = 0$$

This is the requested ODE; $\ddot{\nu}(t)$ is a known function.

Its integral (not requested, but immediate). Integrate from 0 to t; the antiderivative of $\ddot{\theta}+\ddot{\nu}$ is $\dot{\theta}+\dot{\nu}$, and Barrow's rule gives

$$\int_0^t \left(\ddot{\theta}+\ddot{\nu}\right)dt' = \Big[\dot{\theta}+\dot{\nu}\Big]_0^t = \dot{\theta}(t)+\dot{\nu}(t) - \dot{\theta}(0) - \dot{\nu}(0) = 0 \;\Rightarrow\; \dot{\theta}+\dot{\nu} = \Omega_c \equiv \dot{\theta}(0)+\dot{\nu}(0)$$

Integrating once more, $[\theta+\nu]_0^t = \Omega_c t$, so $\theta(t) = \theta(0) + \nu(0) - \nu(t) + \Omega_c t$.

(f) Tension on the cord

From the $\mathbf{e}_c$ component:

$$T = mL\left(\dot{\theta}^2 + 2\dot{\theta}\dot{\nu} + \dot{\nu}^2\right) = mL\left(\dot{\theta}+\dot{\nu}\right)^2$$

With $\theta(t)$ known (and $\nu(t)$ given), $T(t) = mL\left(\dot\theta(t) + \dot\nu(t)\right)^2$; by the integral of (e) it is in fact constant, $T = mL\,\Omega_c^2$. The cord is taut ($T > 0$) as long as $\Omega_c \neq 0$.

General form (as in the official key). Keeping gravity and $\mathbf{a}_0^S$ separate, with $\mathbf{g}_P = \mathbf{g}_S = -\dfrac{\mu_F}{r^2}\mathbf{i}_1$, the $\mathbf{e}_c$ component reads $-mL\dot\theta^2 = -T + m(\mathbf{g}_S - \mathbf{a}_0^S)\cdot\mathbf{e}_c + mL\dot\nu^2 + 2mL\dot\nu\dot\theta$. With $\mathbf{i}_1\cdot\mathbf{e}_c = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_c = \sin\theta$ (the legs of the cord triangle divided by L):

$$T = mL\dot\theta^2 - \frac{\mu_F m\cos\theta}{r^2} + mL\dot\nu^2 + 2mL\dot\theta\dot\nu + m\left(r\dot\nu^2 - \ddot r\right)\cos\theta - m\left(2\dot r\dot\nu + r\ddot\nu\right)\sin\theta$$

For the Keplerian orbit of S, $\ddot r - r\dot\nu^2 = -\mu_F/r^2$ and $2\dot r\dot\nu + r\ddot\nu = \frac{1}{r}\frac{d}{dt}\left(r^2\dot\nu\right) = 0$ (Topic 4: conservation of $h = r^2\dot\nu$ and $p = h^2/\mu_F$), so the second and fifth terms cancel each other and the sixth vanishes, and this expression reduces to $T = mL(\dot\theta + \dot\nu)^2$.

Checks

- **Meaning of $\dot\theta + \dot\nu = \text{const}$.** $\theta + \nu$ is the angle of the cord measured from the inertial axis Fx (shown in the change-of-basis section with the addition formulas). With equal gravity on S and P, both fall together and the astronaut sees no net gravity: relative to a non-rotating frame attached to S he simply turns around S at the constant rate Ω_c, and the cord supplies the centripetal force $mL\Omega_c^2$. In the rotating frame Sx_1y_1 this appears as the balance between the Euler force (which drives $\ddot\theta = -\ddot\nu$), and the centrifugal and Coriolis forces (which, together with $mL\dot\theta^2$, add up to the tension).
- **Limiting cases.** Circular orbit ($e = 0$): $\dot\nu$ is constant, $\ddot\nu = 0$, the Euler force vanishes and $\dot\theta$ is constant. If the astronaut does not rotate in inertial space ($\Omega_c = 0$), the tension vanishes and the taut-cord hypothesis is lost: he would drift freely at constant relative position in a non-rotating frame.
- **Orthogonality:** $\mathbf{i}_1\cdot\mathbf{j}_1 = -\cos\nu\sin\nu + \sin\nu\cos\nu = 0$, $\mathbf{e}_c\cdot\mathbf{e}_\theta = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, all of unit length; $|\mathbf{u}_t| = 1$ by construction.
- **Dimensions:** $[\dot r] = \mathrm{m}\cdot\mathrm{s}^{-1}$ (p in m times $\dot\nu$ in s^{-1}); $[\ddot r] = [r\dot\nu^2] = [r\ddot\nu] = \mathrm{m}\cdot\mathrm{s}^{-2}$; $[mL\ddot\nu] = [mL\dot\nu^2] = [mL\dot\nu\dot\theta] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[\mu_F m/r^2] = \mathrm{m}^3\mathrm{s}^{-2}\cdot\mathrm{kg}/\mathrm{m}^2 = \mathrm{N}$; $\mathbf{u}_t$ is dimensionless.
- **Symbolic check (sympy):** $\mathbf{a}_0^S$ from differentiating $r(\nu(t))\,\mathbf{i}_1$ twice in $\mathcal{B}_0$, the formulas for $\dot r$ and $\ddot r$, the unit tangent, $\mathbf{v}_1^P$, $\mathbf{a}_1^P$, the decomposition of Notes Eq. 3.27, the four inertia forces, $\ddot\theta = -\ddot\nu$, $T = mL(\dot\theta + \dot\nu)^2$, the equality of the general formula with the key, and $\mathbf{a}_0^S = -\mu_F/r^2\,\mathbf{i}_1$ for $\dot\nu = h/r^2$, $p = h^2/\mu_F$: all verified.

- **Check against the official key:** (`Problems.pdf`, PDF pages 73–74) it gives $\mathbf{a}_0^S$, $\dot{r}$, $\ddot{r}$, the Euler force $mL\ddot{\nu}(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1)$ and the general tension formula written above; all of them coincide with the results of this solution, and the general formula reduces to $mL(\dot{\theta}+\dot{\nu})^2$ for the Keplerian orbit.

Source: Problems.pdf (PDF page 13); key on PDF pages 73-74